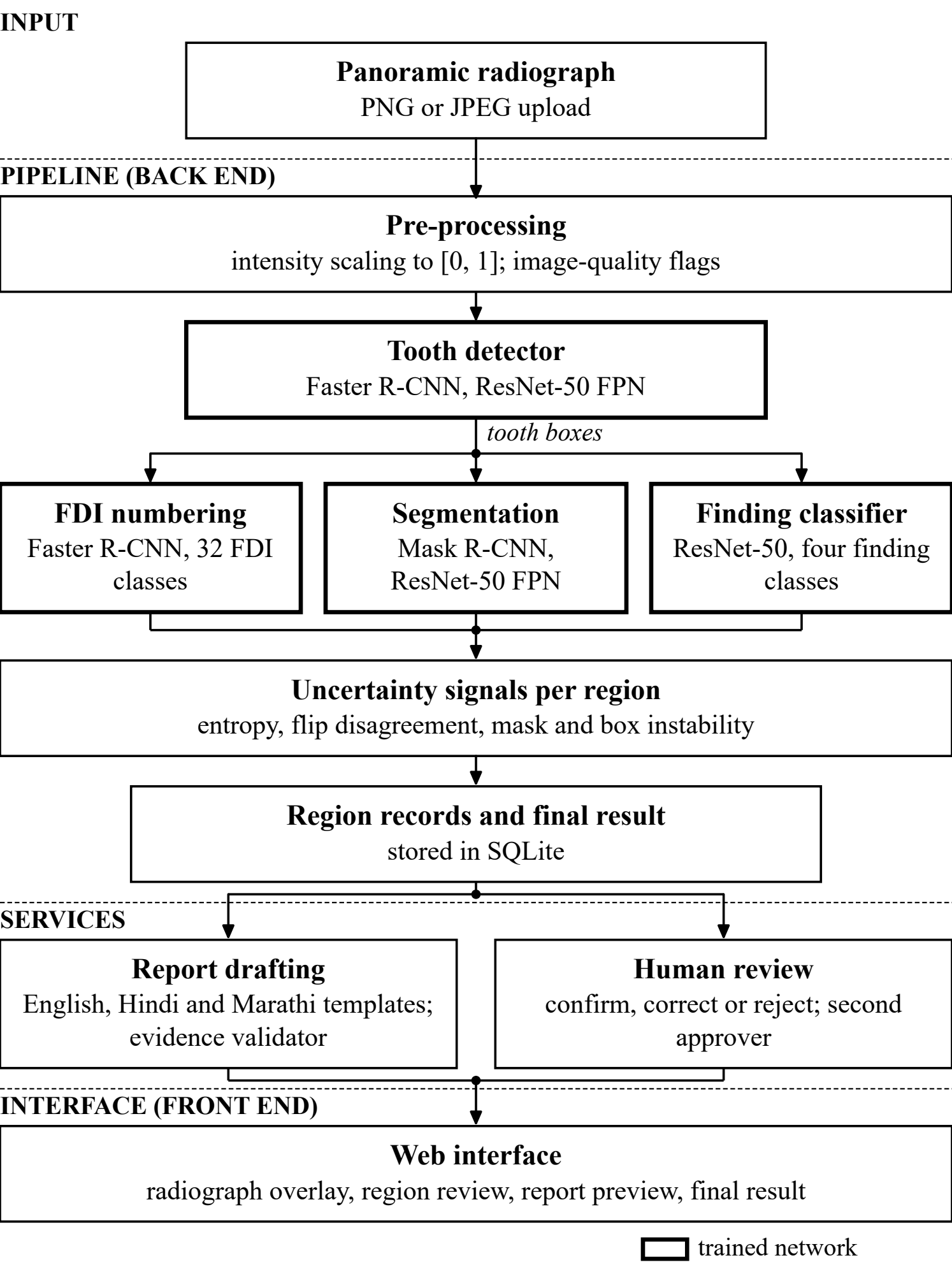

INPUT
Panoramic radiograph
PNG or JPEG upload
PIPELINE (BACK END)
Pre-processing
intensity scaling to [0, 1]; image-quality flags
Tooth detector
Faster R-CNN, ResNet-50 FPN
tooth boxes
FDI numbering
Faster R-CNN, 32 FDI classes
Segmentation
Mask R-CNN, ResNet-50 FPN
Finding classifier
ResNet-50, four finding classes
Uncertainty signals per region
entropy, flip disagreement, mask and box instability
Region records and final result
stored in SQLite
SERVICES
Report drafting
English, Hindi and Marathi templates; evidence validator
Human review
confirm, correct or reject; second approver
INTERFACE (FRONT END)
Web interface
radiograph overlay, region review, report preview, final result
trained network